

Mohammad Reza Gerami

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Summary

Cybersecurity engineer with 15 years of experience across network security, cloud infrastructure, and applied cryptography. Currently completing a PhD in post-quantum cryptography (PQC) at Polytechnique Montréal, with hands-on experience implementing NIST-standardized algorithms (ML-KEM, ML-DSA) and hybrid encryption schemes for IoT and critical infrastructure environments. Combines research-level cryptographic knowledge with production experience in cloud security, SIEM, and DevSecOps automation.

Core Skills

Cryptography & Security: Post-Quantum Cryptography (NIST FIPS 203/204/205), Hybrid Encryption, Zero Trust Architecture, IoT/UAV Network Security, ISO/IEC 27001

Cloud & DevOps: AWS, Azure, OpenStack, Docker, Kubernetes, Ansible, CI/CD, Linux

Network Security: SIEM (Splunk ES, ELK), Vulnerability Assessment (Nessus), SDN, Firewalls, IDS/IPS

Programming: Python (security automation), Shell scripting, C/C++

Professional Experience

PhD Researcher — Cybersecurity & Post-Quantum Cryptography Dec 2022 – Present
Polytechnique Montréal, Montreal, QC

- Designing hybrid post-quantum cryptographic protocols for IoT-enabled UAV network security.
- Applying federated learning and deep reinforcement learning to improve resilience of distributed drone networks.
- Built a next-generation SIEM implementation (Splunk ES, ELK) with custom detection rules, reducing incident resolution time by 20%.
- Ran anomaly detection on 4.7 million supercomputer log records using the ELK stack, cutting detection time by 30%.
- Simulated UAV swarm security scenarios using STM Cortex-M4 hardware and the NS-3 network simulator.

Founder & Lead Architect Nov 2025 – Present
QuantumGuard Bridge (QGB), Montreal, QC — part-time alongside PhD research

- Building a platform to help organizations inventory legacy cryptography and plan migration to NIST-approved post-quantum algorithms.
- Designing hybrid cryptographic schemes that preserve backward compatibility during the transition period.
- Leading early market validation and technical scoping for the minimum viable product.

Cloud Security & SDN Engineer Jul 2021 – Nov 2022
Sadad Informatics Corporation, Tehran, Iran

- Automated cloud deployment workflows using Ansible and Python-based CI/CD, reducing deployment time by 70%.
- Deployed monitoring stacks (Zabbix, SolarWinds, ELK), improving system uptime by 40%.
- Designed an SDN architecture with control/data-plane separation, improving network efficiency by 25%.

Co-Founder & CTO

Sept 2019 – Jul 2021

Vira Security Solutions, Tehran, Iran

- Redesigned corporate network architecture, reducing infrastructure costs by 40%.
- Delivered a security-awareness program for 150+ employees, reducing internal security incidents by 40%.
- Hardened Linux/Windows servers and implemented physical access controls, reducing breaches by 30%.

Network & System Administrator

Sept 2017 – Sept 2019

Research Institute for Endocrine Sciences, Tehran, Iran

- Designed secure high-performance infrastructure for genetic data processing, improving processing speed by 85%.
- Managed over 400TB of genomic data with high-availability architecture.
- Scaled VDI (VMware Horizon) access from 5 to 50 concurrent researchers.

Network Security Engineer

Jun 2014 – Sept 2017

Farazpajohan, Tehran, Iran

- Led vulnerability assessments using Nessus, remediating 90% of identified critical vulnerabilities.
- Implemented ISO/IEC 27001-aligned security policies across virtualized environments (VMware ESXi, Citrix).

HPC & Linux Engineer

Jun 2010 – May 2014

Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

- Deployed and optimized over 30 HPC clusters using CentOS and Scientific Linux.
- Reduced computation time by 65% via InfiniBand communication and GPU integration.

Education

Ph.D., Computer Engineering — Polytechnique Montréal	Expected 2027
M.Sc., Computer Networks — Amirkabir University of Technology	2013
B.Sc., Software Engineering — Shamsipour Technical College	2009

Certifications

Cisco CCNP Security | LPIC-3 | VMware VCP | Red Hat Certified System Administrator (Open-Stack) | MCSE | MCITP | IBM Data Science Professional

Awards

Winner, Polytechnique Montréal Cybersecurity Innovation Competition (2025)